TEXORA · COURSE NOTES · LECTURE SHEET

Drawing Equipment and the Use of Instruments

The drawing board kit, instrument by instrument — what each tool is, how it is correctly used, the standard sheet sizes, and the line types every instrument must produce.

ME-1102 · ENGINEERING DRAWING-I · TOPIC 2 · LEVEL 1 TERM 1

1. The sheet itself: sizes and layout

Drawing paper comes in the ISO A series, each size half the previous (folded along its length), all with sides in the ratio 1 : √2:

SIZE	DIMENSIONS (MM)	TYPICAL USE
A0	841 × 1189	Plant layouts, large assemblies
A1	594 × 841	General engineering drawings
A2	420 × 594	College drawing sheets (the course standard)
A3	297 × 420	Practice, sketches, prints
A4	210 × 297	Office copies, reports

A working sheet is laid out with a **border** (a rectangle typically 10 mm inside the edges), a wider **filing margin** (about 20 mm on the left edge) so the sheet can be filed without eating the drawing, and the **title block** at bottom-right (Topic 1). The sheet is fixed to the board with clips or tape at the corners — never pins through the middle, never folded.

2. The instruments, one by one

INSTRUMENT	WHAT IT IS	CORRECT USE
Drawing board	A flat, seasoned-wood (or laminated) board with one true working edge, sized to the sheet (boards are designated D0–D3 to match sheets)	Sheet aligned to the working edge; the T-square rides this edge only — a damaged edge ruins every line on the sheet

T-square	A long blade fixed at right angles to a head (stock) that slides along the board's edge	Draws horizontal lines (along its upper edge); verticals and angles come from set squares resting on it. Press the head firmly to the edge; draw left to right (right-handed)
Set squares	Two transparent triangles: 45°/45°/90° and 30°/60°/90°	Used against the T-square for verticals and for 15°, 30°, 45°, 60°, 75° and combined angles; slide, don't lift, to step parallel lines
Mini-drafter	An arm-and-protractor head combining T-square, set squares, protractor and scale on one mount	Clamped to the board's corner; the head is locked at 0° for horizontals/verticals or set to any angle — the modern replacement for the separate tools on course sheets
Compass	Two legs — needle point and pencil (or pen) point — hinged at the top; large and small (bow) sizes	Circles and arcs. Bend both legs so they stand perpendicular to the paper at the working radius; lean the compass slightly into the direction of travel; one smooth sweep, clockwise
Divider	Two needle legs, no pencil	Transferring and stepping off equal distances, dividing lines, picking dimensions from a scale — it measures; it never draws
Protractor	A semicircular (or circular) graduated scale, 0–180°	Setting out and measuring angles not covered by the set squares
Scale (ruler)	A flat or triangular rule carrying several graduated scales (1:1, 1:2, 1:5, 1:10 …), i.e. representative fractions	Reading and laying off dimensions at reduced/enlarged scale; never use its edge as a straightedge for lines — that is the T-square's job, and scale edges bruise
French curves	Flat templates of irregular curves	Drawing smooth curves through plotted points that no compass arc fits; match a section, draw, re-match — in short overlapping portions
Pencils	Graded 9H (hardest) to 7B (softest); drawing work lives in H–3H	2H/HB range by task (§3); keep points conical for lines, chisel for long straight runs; rotate while drawing to wear evenly

Eraser & shield, brush	Soft eraser; a thin metal shield with slots; a dusting brush	Erase through the shield's slot to protect neighbouring lines; brush (never blow or wipe by hand — graphite smears and hands grease the sheet)

3. Pencil grades by task

TASK	GRADE	WHY
Construction lines, guidelines, centre lines	2H or 3H	Hard lead gives fine, light lines that guide without shouting — and erase cleanly
Object (visible) lines, final outlines	H or HB	Softer lead gives the dense black the finished outline needs
Lettering	H (kept sharp)	Uniform thin strokes (Topic 1)
Freehand sketching	HB or B	Soft lead responds to pressure in ideation work

4. The lines the instruments must produce

Instruments exist to serve the standard **alphabet of lines**; every line type is a word in the drawing's language:

LINE TYPE	APPEARANCE	MEANING / USE
Continuous thick	Solid, ~0.5–0.7 mm	Visible outlines and edges of the object
Continuous thin	Solid, ~0.2–0.3 mm	Dimension, extension and leader lines; hatching; construction
Dashed (hidden)	Short dashes, thin	Edges hidden behind the object's surfaces
Chain thin (centre)	Long dash–short dash	Centre lines of circles and symmetrical parts; axes
Chain thin (cutting plane)	Chain, thickened at ends	Where an imaginary cutting plane passes (sections)
Freehand thin	Wavy solid	Break lines — where a long part is shortened on paper

5. Care and accuracy habits

- Instruments are precision tools: a compass with a bent needle leg draws spirals, not circles; check hinges for play before the examination sheet, not during it.
- Keep the sheet clean — a sleeve or a cover sheet under the drawing hand; sharpen pencils away from the sheet.
- Draw construction first, light and complete; finish with object lines in order — curves before straights where they join, so tangencies are exact.
- Every length is taken from the scale or stepped with the divider from one datum — never chained from the last measurement (errors compound; Topic 4's projections inherit them).

6. Glossary — key terms of this topic

TERM	MEANING
Representative fraction (RF)	The ratio of a length on the drawing to the real length (1:2 means RF = 1/2).
Set squares	The 45° and 30°/60° triangles used with the T-square for verticals and standard angles.
Bow compass/divider	Small spring-hinged compass/divider for small circles and fine stepping.
Mini-drafter	A combined protractor-scale head on an articulated arm, replacing T-square and set squares.
Hidden line	The dashed line showing an edge concealed by the object itself.
Centre line	The long–short chain line marking axes and centres of symmetry.
Filing margin	The extra-wide left margin (≈20 mm) allowing the sheet to be filed without loss.

7. Key points and summary

- Sheets follow the ISO A series (course standard A2) with border, filing margin and title block; the board's true edge anchors everything.
- Each instrument has one job: T-square horizontals, set squares angles and verticals, compass circles, divider measuring, scale reading RF scales, French curves irregular curves — with pencil grades matched to line duties (2H–3H construction, H–HB outlines).
- The alphabet of lines — thick/thin continuous, dashed hidden, chain centre — is what the whole kit exists to produce consistently.

8. Practice

Written and drawing exercises; brief model answers follow.

1. List the A-series sheet sizes A0–A4 with dimensions, and state the side ratio common to all.
2. Assign one correct task to each: T-square, 45° set square, divider, French curve, protractor — and name one misuse of each to avoid.
3. Choose pencil grades for: construction lines, final outline of a machine part, lettering, and a freehand sketch — with one-line reasons.
4. Draw (describe in words) the six standard line types of §4 and give one use of each on a simple bracket drawing.
5. A drawing is made at 1:5. Write its representative fraction, and the real length of a part measuring 24 mm on paper.
6. Drawing exercise: with a mini-drafter or set squares, construct lines at 15°, 30°, 45°, 60°, 75° and 90° to a base line, labelling each — using combinations of the two set squares where needed.
7. Why is a scale never used as a straightedge, and a divider never used to draw?
8. A compass draws slightly oval circles. Give the two most likely instrument faults and how each is checked.

Answer key

1. A0 841×1189, A1 594×841, A2 420×594, A3 297×420, A4 210×297 mm; ratio 1 : √2 throughout (each size is the previous halved lengthwise).
2. T-square — horizontals (misuse: using the blade’s damaged lower edge or drawing verticals with it). 45° set square — 45°/vertical lines against the T-square (misuse: using it freehand without a base edge). Divider — transferring/stepping distances (misuse: drawing arcs with it — no pencil leg). French curve — irregular curves through points (misuse: forcing one long match instead of short sections). Protractor — odd angles (misuse: reading the wrong of its two scales).
3. Construction: 2H–3H (light, fine). Outline: H–HB (dense black). Lettering: H sharp (uniform stroke). Sketch: HB–B (responsive, dark).
4. Continuous thick — the bracket’s visible outline; continuous thin — its dimension lines and hatching; dashed — the hidden far edge of its hole; chain thin — the hole’s centre lines; chain with thick ends — the cutting plane for a section view; freehand wavy — a break where the bracket’s long arm is shortened.
5. RF = 1/5. Real length = 24 × 5 = 120 mm.
6. Self-check: 45° and 90° from either triangle; 30°/60° from the other; 15° = 45° − 30° and 75° = 45° + 30° by combining both set squares on the T-square.
7. The scale’s graduated edges are measuring surfaces — drawing along them wears and bruises the graduations and the edge is not guaranteed true; the divider has two needle points and no marking

leg, so it can only step and transfer distances — drawing with it scores the sheet.

8. (1) A bent or loose needle leg — check by closing the legs: points should meet exactly, and the hinge should have no side play. (2) Legs not bent perpendicular at the working radius — check on paper: at the set radius both points must stand vertical to the sheet; if the pencil leg leans, the radius changes through the sweep, giving the oval.
